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SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

MATHEMATICS P2

MAY/JUNE 2025

MARKS: 150

TIME: 3 hours

This question paper consists of 13 pages and 1 information sheet.

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of 10 questions.
2. Answer ALL the questions in the SPECIAL ANSWER BOOK provided.
3. Clearly show ALL calculations, diagrams, graphs, etc. which you have used in determining your answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. An information sheet with formulae is included at the end of the question paper.
9. Write neatly and legibly. .

QUESTION 1

An insurance broker signed contracts with 15 people. The monthly premium (in rands) payable on each contract is given below.

134	215	325	326	362	429	515	531	598	610	624	728	923	1 034	1 200
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- 1.1 Calculate the mean of the data. (2)
- 1.2 Write down the standard deviation of the data. (1)
- 1.3 Calculate how many monthly premiums are within ONE standard deviation of the mean. (2)
- 1.4 The insurance company decided to increase the monthly premiums.
- Monthly premiums that were less than R500 increased by 18%
 - Monthly premiums that were equal to or more than R500 increased by $k\%$

After these increases were applied to the above data, the new mean monthly premium was R686,44. Calculate the value of k .

(4)
[9]

QUESTION 2

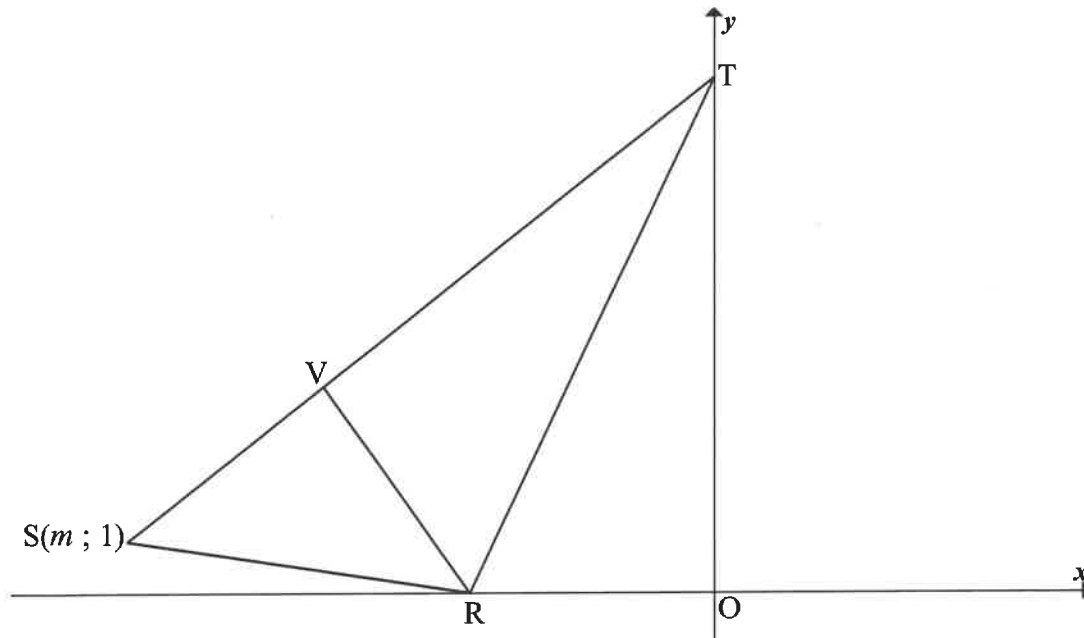
The manager of a supermarket decided to do a survey on the number of items that a customer ordered online and the time (in minutes) that a packer took to have the order ready for delivery. The supermarket received 10 online orders on a certain day. The information for these 10 orders is shown in the table below.

Number of items (x)	10	3	20	14	17	9	12	18	15	19
Time (in minutes) (y)	5	5	9	7	6	6	8	11	10	12

- 2.1 Draw a scatter plot on the grid provided in the ANSWER BOOK. (3)
- 2.2 Determine the equation of the least squares regression line. (3)
- 2.3 Write down the correlation coefficient of the data. (1)
- 2.4 The supermarket received an online order for 13 items. Predict how long (in minutes) it will take a packer to pack the order and have it ready for delivery. (2)
- 2.5 Explain why the y -intercept of the least squares regression line in QUESTION 2.2 does NOT make sense in this context. (1)
- [10]**

QUESTION 3

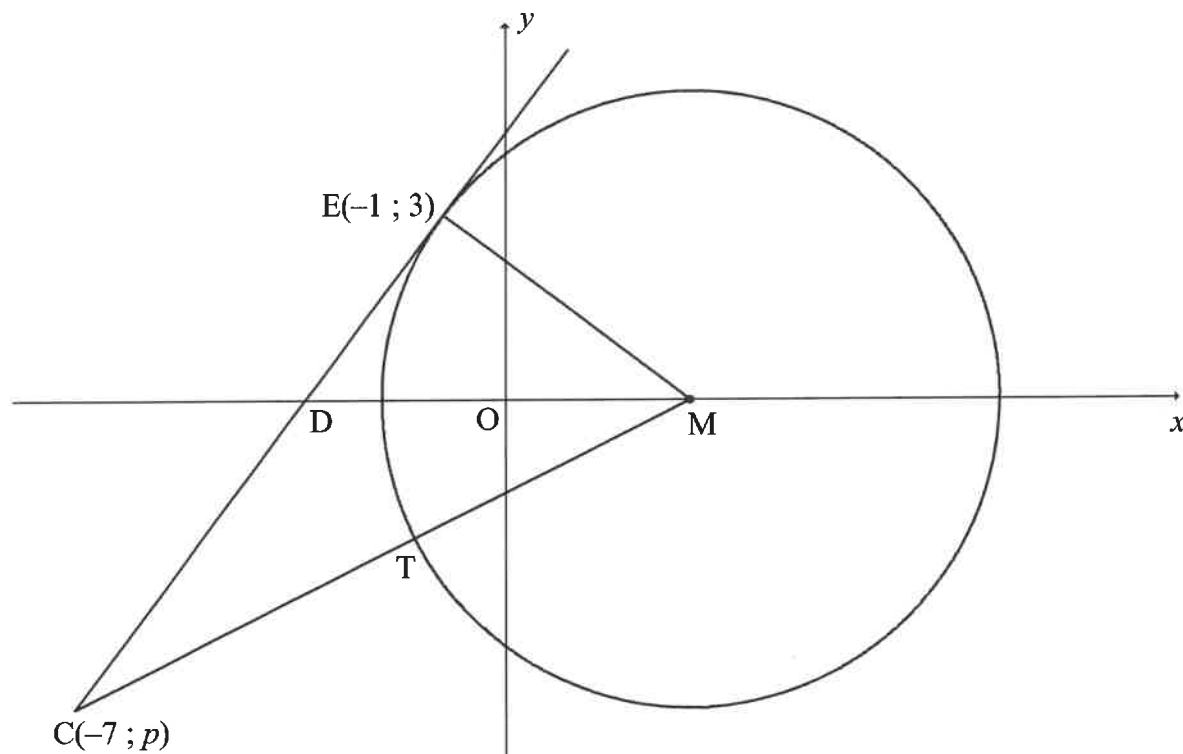
In the diagram below, $\triangle SRT$ is drawn where R lies on the x -axis and S lies to the left of R . T lies on the y -axis and the coordinates of S are $(m ; 1)$. The equation of RT is $2x - y + 10 = 0$.



- 3.1 Calculate the coordinates of R . (2)
 - 3.2 Calculate the length of RT . Leave your answer in surd form. (3)
 - 3.3 If it is also given that $2RT^2 = 5SR^2$, calculate the value of m . (4)
 - 3.4 It is further given that V lies on ST such that VR is perpendicular to ST . Determine the equation of VR in the form $y = mx + c$. (5)
 - 3.5 Hence, show that the coordinates of V are $(-8 ; 4)$. (2)
 - 3.6 If R' is the reflection of R about the line $x = 0$, calculate the area of $RVTR'$. (5)
- [21]**

QUESTION 4

In the diagram, M is the centre of the circle having equation $(x-3)^2 + y^2 = 25$. $E(-1; 3)$ and T are points on the circle. EC is a tangent to the circle at E and cuts the x -axis at D . $ED = \frac{15}{4}$ units. MT is produced to meet the tangent at $C(-7; p)$.



- 4.1 Write down the size of $\hat{CEM}$. (1)
- 4.2 Determine the equation of the tangent EC in the form $y = mx + c$. (4)
- 4.3 Calculate the length of DM . (3)
- 4.4 Show that $p = -5$. (1)
- 4.5 Calculate the coordinates of S if $SEMC$ is a parallelogram and $x_s < 0$. (3)
- 4.6 If the radius of the circle, centred at M , is increased by 7 units, determine whether S lies inside or outside the new circle. Support your answer with the necessary calculations. (3)
- 4.7 If ET is drawn, calculate the size of $\hat{ETM}$. (5)

[20]

QUESTION 5

5.1 If $\cos \theta = -\frac{5}{13}$ where $180^\circ < \theta < 360^\circ$, determine, **without using a calculator**, the value of:

5.1.1 $\sin^2 \theta$ (3)

5.1.2 $\tan(360^\circ - \theta)$ (2)

5.1.3 $\cos(\theta - 135^\circ)$ (4)

5.2 Simplify the expression to a single trigonometric term: $\frac{2 \cos(180^\circ - x) \sin(-x)}{1 - 2 \cos^2(90^\circ - x)}$ (6)

5.3 Calculate the value of the following expression **without using a calculator**:
 $(\tan 92^\circ)(\tan 94^\circ)(\tan 96^\circ) \dots (\tan 176^\circ)(\tan 178^\circ)$ (4)
[19]

QUESTION 6

6.1 Prove that $2 \cos^2(45^\circ + x) = 1 - \sin 2x$. (4)

6.2 Consider the expression: $\sin(A - B) - \sin(A + B)$

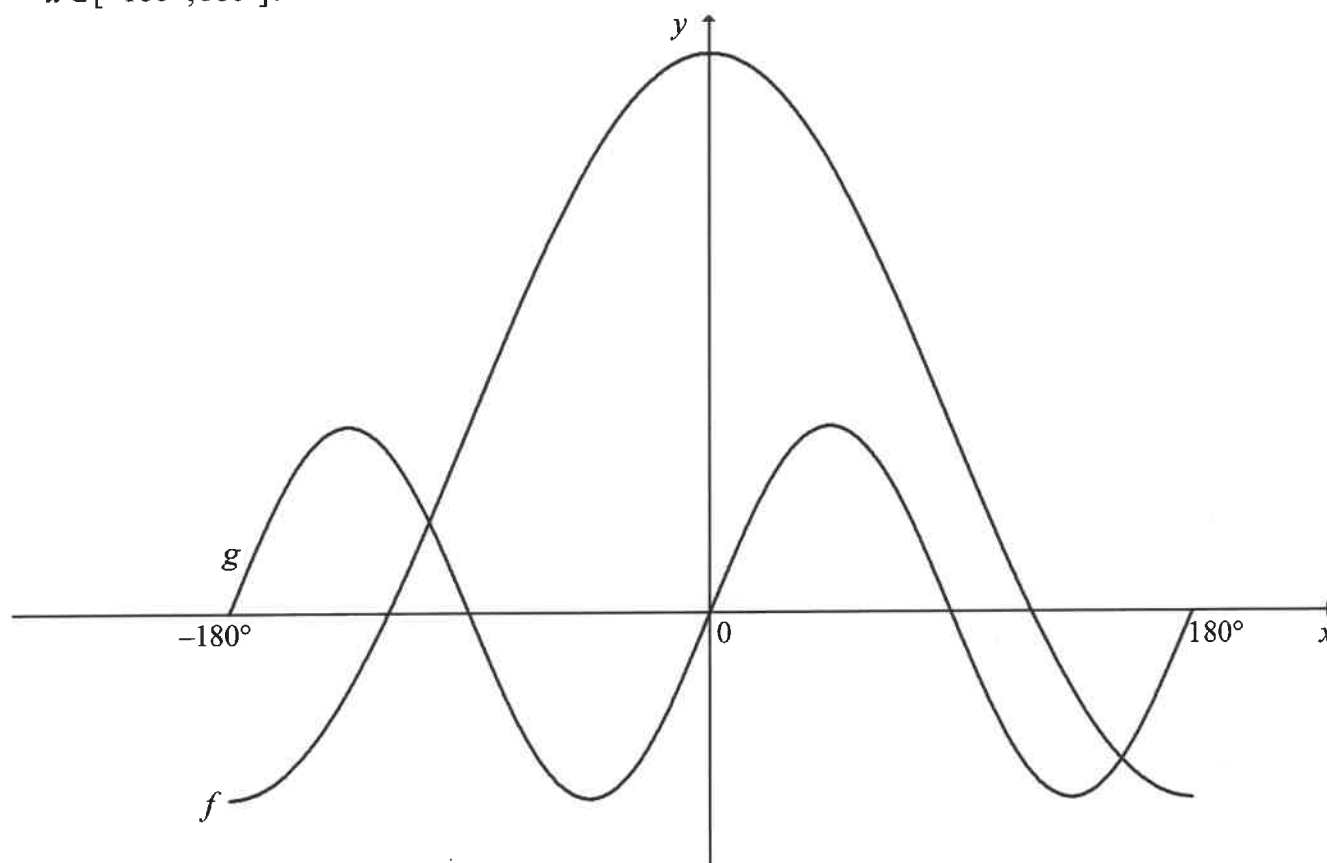
6.2.1 Prove that $\sin(A - B) - \sin(A + B) = -2 \cos A \sin B$. (2)

6.2.2 Simplify the following expression to a single term: $\sin 4x - \sin 10x$ (2)

6.2.3 Hence, determine the solution for $\sin 4x - \sin 10x = \sin 3x$ for $x \in [0^\circ; 30^\circ]$. (5)
[13]

QUESTION 7

In the diagram, the graphs of $f(x) = 2 \cos x + 1$ and $g(x) = \sin 2x$ are drawn for the interval $x \in [-180^\circ; 180^\circ]$.



7.1 Write down the range of f . (1)

7.2 Write down the period of g . (1)

7.3 For which values of x , in the interval $x \in [-180^\circ; 180^\circ]$, is f increasing? (1)

7.4 Use the graphs to determine the values of x , in the interval $x \in [-180^\circ; 180^\circ]$, for which:

7.4.1 $g(x) \cdot f'(x) < 0$ (2)

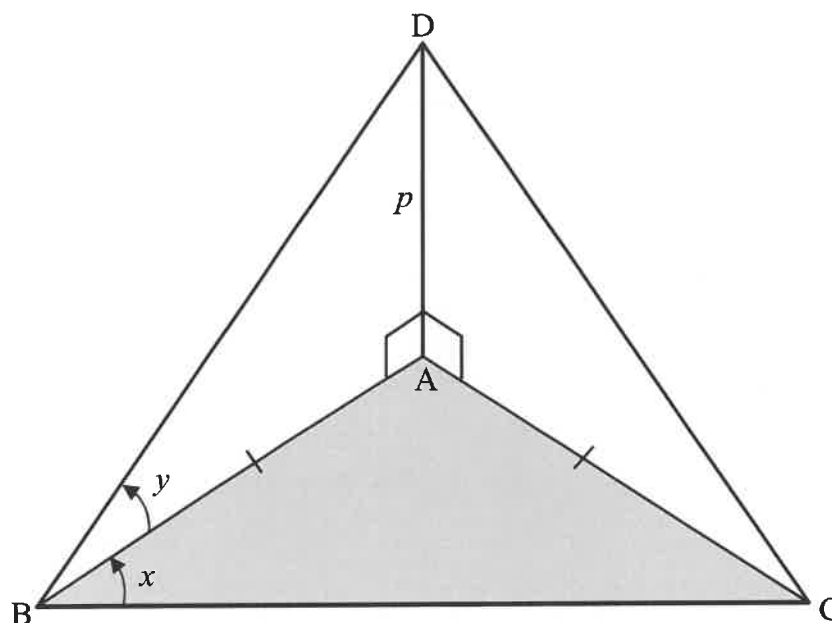
7.4.2 $\cos x \leq -\frac{1}{2}$ (3)

7.5 Graph g is shifted 45° to the right to obtain a new graph h . Determine the equation of h in its simplest form. (2)

[10]

QUESTION 8

In the diagram, A, B and C lie in the same horizontal plane with $AB = AC$. D is directly above A such that $2AD = BC$. Also, $AD = p$, $\hat{ABC} = x$ and $\hat{DBA} = y$.

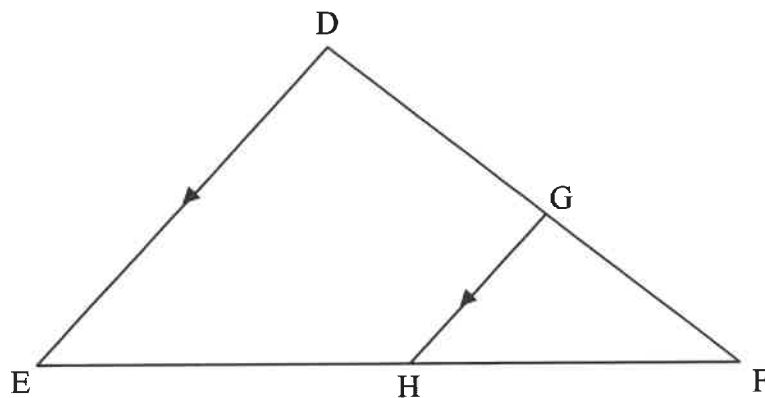


- 8.1 Determine AB in terms of p and y . (2)
- 8.2 Show that $\cos x = \tan y$. (4)
- 8.3 If $x = 60^\circ$, calculate the size of y . (2)
- [8]**

Provide reasons for your statements in QUESTIONS 9 and 10.

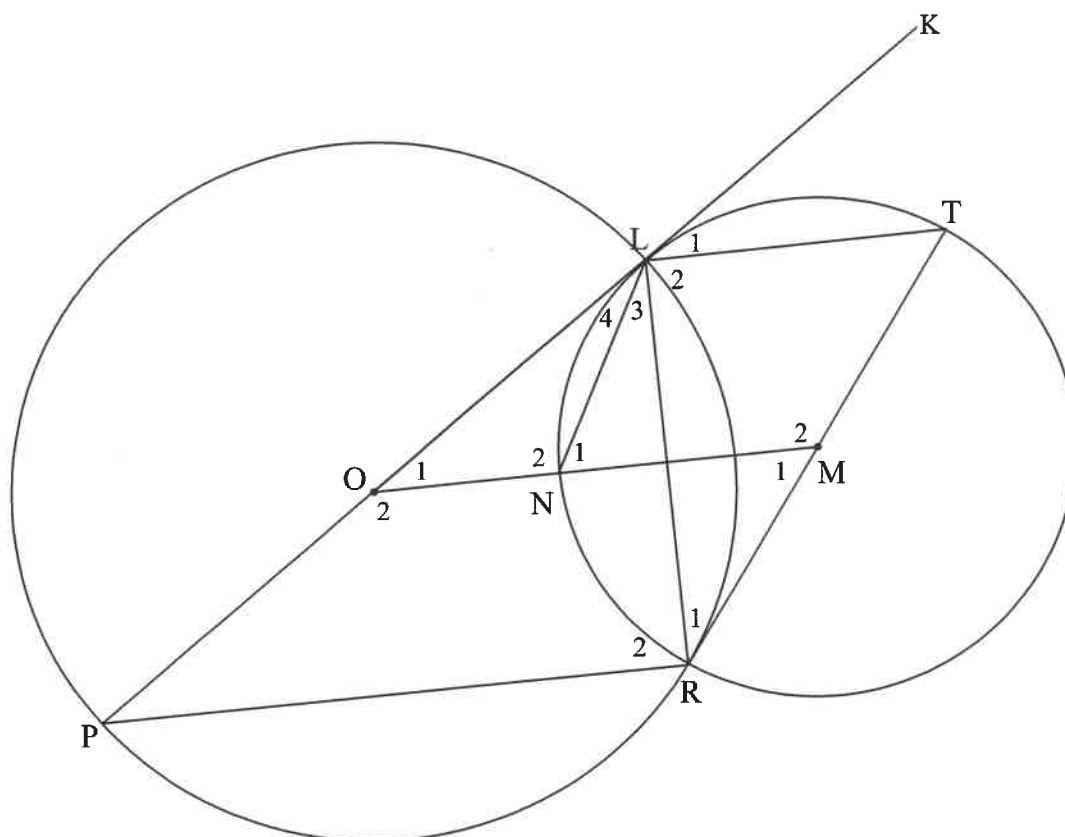
QUESTION 9

- 9.1 In the diagram, $\triangle DEF$ is drawn. Line GH intersects DF and EF at G and H respectively such that $GH \parallel DE$ and $\frac{GF}{DG} = \frac{2}{5}$.



- 9.1.1 Write down, with a reason, the value of $\frac{HF}{EH}$. (2)
- 9.1.2 If $EF = 21$ cm, calculate the length of EH . (2)
- 9.1.3 Write down a triangle which is similar to $\triangle FGH$. (1)
- 9.1.4 Hence, calculate the value of $\frac{GH}{DE}$. (2)

- 9.2 In the diagram, POL is a diameter of the larger circle with centre O . TMR is a diameter of the smaller circle with centre M . The two circles intersect at L and R . PLK is a tangent to the smaller circle at L and TR is a tangent to the larger circle at R . OM intersects the smaller circle at N . Straight lines LT , LR , LN and PR are drawn.

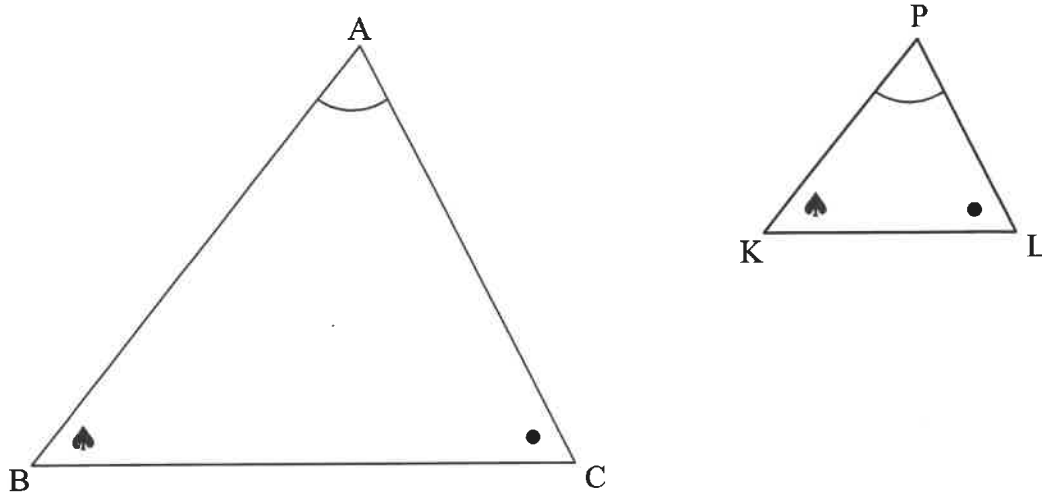


Prove, giving reasons, that:

- 9.2.1 $LT \parallel PR$ (4)
- 9.2.2 $LORM$ is a cyclic quadrilateral, if it is also given that $LT \parallel OM$ (5)
- 9.2.3 LN bisects OLR (4)
- [20]

QUESTION 10

10.1 In the diagram, $\triangle ABC$ and $\triangle PKL$ are drawn such that $\hat{A} = \hat{P}$, $\hat{B} = \hat{K}$ and $\hat{C} = \hat{L}$.

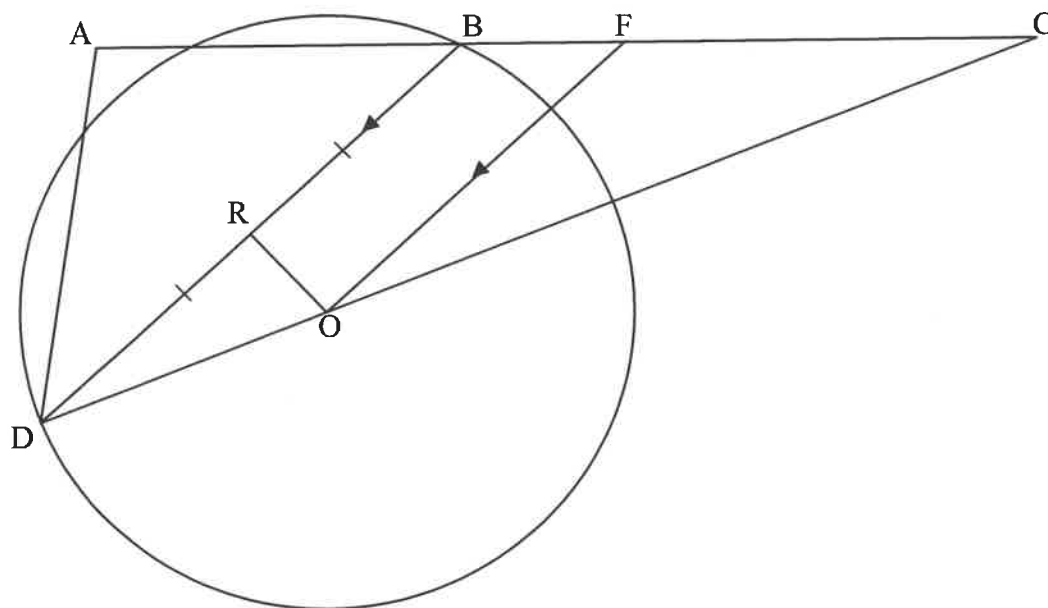


Use the diagram in the ANSWER BOOK to prove the theorem which states that if two triangles are equiangular, then the corresponding sides are in proportion,

i.e. that $\frac{AB}{PK} = \frac{AC}{PL}$.

(6)

- 10.2 In the diagram, O is the centre of the circle. Points D and B lie on the circle. Points A and C lie outside the circle such that side AC of $\triangle ADC$ passes through B . F is a point on BC such that $FO \parallel BD$. $DR = RB$ and RO is drawn.



10.2.1 Prove, with reasons, that $\triangle CFO \parallel \triangle CBD$. (3)

10.2.2 If it is given that $\hat{RDO} = \hat{FCO}$, show, with reasons, that $OF \cdot CD = CO \cdot BC$ (2)

10.2.3 It is further given that $DC = 19,2$ units, $BD = 12$ units and $\frac{RO}{RD} = \frac{3}{4}$
Prove, with reasons, that $BF = \frac{75}{16}$ (6)

10.2.4 Calculate the size of $\hat{ABD}$. (3)
[20]

TOTAL: 150

INFORMATION SHEET: MATHEMATICS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}; r \neq 1$$

$$S_\infty = \frac{a}{1 - r}; -1 < r < 1$$

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x - a)^2 + (y - b)^2 = r^2$$

In $\triangle ABC$:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2} ab \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2 \sin^2 \alpha \\ 2 \cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$